Attribute-Contrastive Classifier-Free Guidance for Disentangled Conditional Diffusion

Replacing CFG's Null Baseline with Targeted Anchor Conditions

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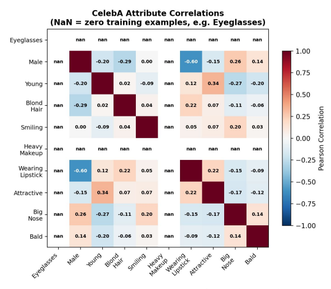
Background

Introduction:
Diffusion models generate images by learning to reverse a gradual noising process. Classifier-Free Guidance (CFG) enables conditional generation by combining two noise predictions at each step: one conditioned on the target attributes, one unconditional:

ε_guided = ε_uncond + w · (ε_cond − ε_uncond)

The scale `w` controls conditioning strength.
Problem: the guidance direction `ε_cond − ε_uncond` captures everything different about the condition — including spurious correlations in the training data.

Data:
- CelebA — 63,715 face images (64x64), each labeled with 40 binary attributes
- Strong inter-attribute correlations exist (e.g., Smiling ↔ High_Cheekbones)



Base Model: UNet (18.6M params) with FiLM conditioning. Attribute vector mapped through a 2-layer MLP (40→512→512), added to the timestep embedding, and broadcast globally to all ResBlocks. Trained 50k steps on a single RTX 4060 (~5 hrs). DDIM sampling, 100 steps.

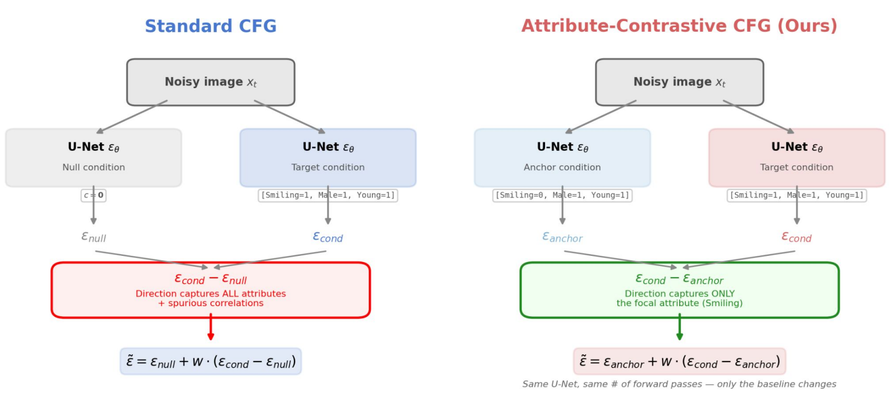
Hypothesis: Replacing the null baseline with a targeted anchor that differs in only the focal attribute will (1) reduce KID, (2) increase sample diversity, and (3) resist mode collapse at high guidance scales because the contrastive direction has smaller L2 norm and isolates the focal attribute's causal effect.

Methods

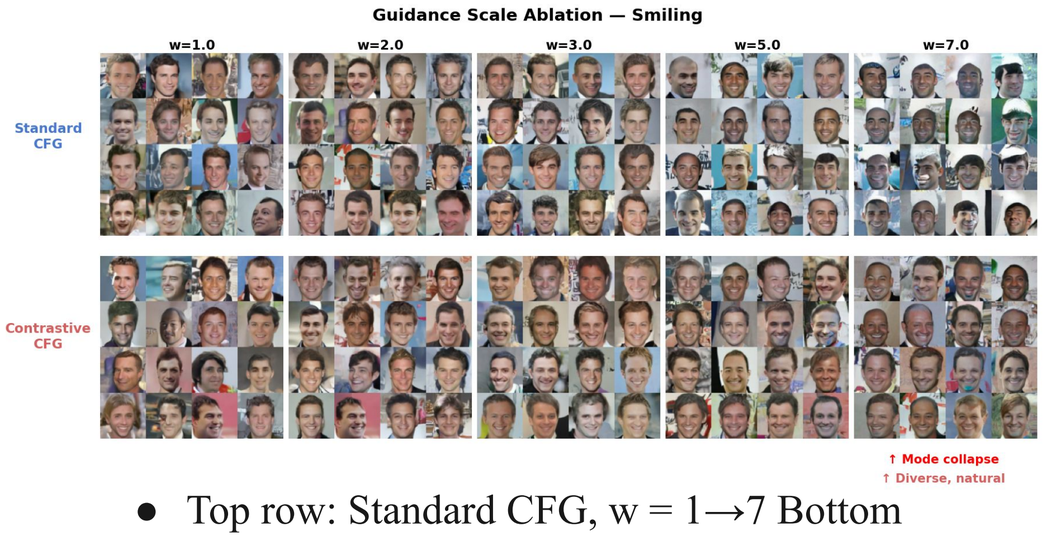
Replace the null baseline with an anchor identical to the target but with the focal attribute flipped:

Target: [Smiling=1, Male=1, Young=1]
Anchor: [Smiling=0, Male=1, Young=1]

ε_guided = ε_anchor + w · (ε_target − ε_anchor)
- last term is isolation of Smiling

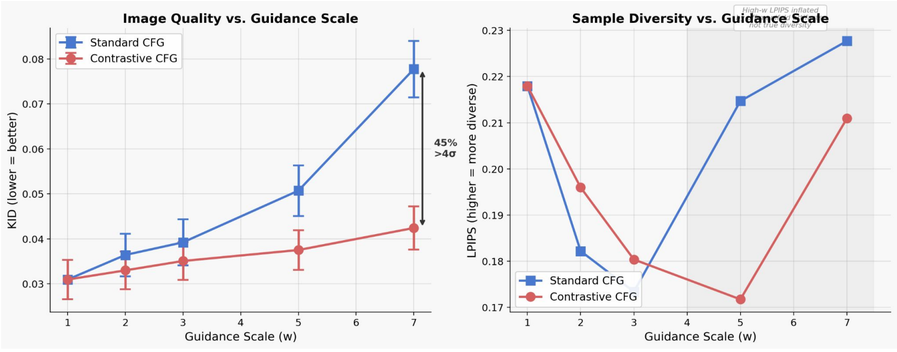


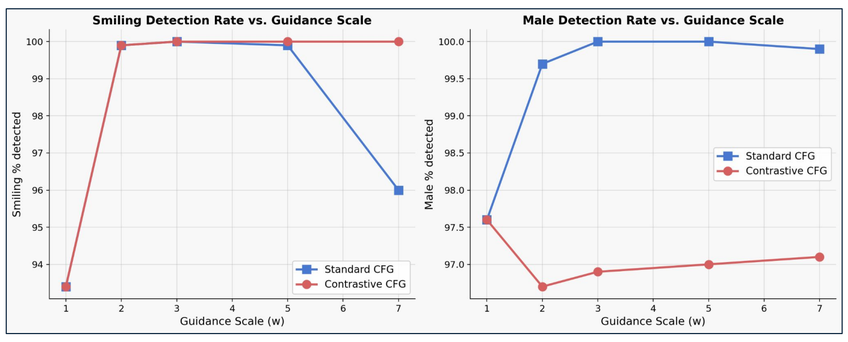
Results



- Top row: Standard CFG, w = 1→7 Bottom row: Contrastive CFG, w = 1→7.
- Standard collapses at w=7, Contrastive stays diverse

Quantitative: KID & Diversity Across Guidance Scales





Quantitative Results Summary

Metric	Standard CFG	Contrastive CFG (Ours)
KID (w=2) ↓	0.0364 ± 0.0047	**0.0330 ± 0.0042 (−9.4%)**
KID (w=7) ↓	0.0777 ± 0.0063	**0.0424 ± 0.0048 (−45.4%)**
KID degrade w=1→7	+151%	**+37%**
LPIPS Diversity ↑	0.182	**0.196 (+7.7%)**
Male % (w=2)	99.7%	**96.7%**
Extra Compute	—	**None**

Smiling sweep: focal attribute = Smiling

Interpretation:
Three independent metrics confirm each prediction from our hypothesis:

1. Quality (KID): Standard CFG degrades 151% from w=1 to w=7 (0.0309→0.0777). Contrastive CFG degrades only 37% (0.0310 →0.0424). At w=7, the gap is > 4 stds with non overlapping error bars. The contrastive direction avoids off-manifold extrapolation.

2. Diversity (LPIPS): Contrastive CFG preserves 7.7% more diversity (0.196 vs 0.182). LPIPS is unreliable at high w because off-manifold artifacts inflate pixel distances.

3. Attribute leakage (classifier): Standard CFG pushes Male detection to 100% at w≥3, over-amplifying a correlated attribute. Contrastive CFG holds ~97% across all scales. Both achieve ~100% Smiling (the target attribute), confirming the difference is in unintended amplification

At w=1, both methods produce identical outputs (KID=0.0309, LPIPS=0.218), confirming all differences come from the guidance direction.

Conclusion

Reasoning:
The anchor is semantically close to the target (differs by one bit). The guidance direction has smaller L2 norm, so the same `w` produces less off-manifold extrapolation. Explains both the mode collapse resistance and the flatter KID curve.

Limitations:
- Our CelebA subset has zero positive examples for several attributes (Eyeglasses, Heavy_Makeup, Wearing_Hat). No guidance method can generate concepts absent from training data
- Requires structured conditioning: Binary attributes allow exact bit-flips. Free-form text requires approximate prompt engineering.

Relevance:
Recent work has begun to question the null baseline: Composable Diffusion [3] showed score functions can be added and subtracted to compose concepts, Wu & De la Torre [2] proposed using contrastive text prompts as baselines instead of null, and Sadat et al. [4] demonstrated that any condition can serve as a baseline. Meanwhile, CFG++ [5] and Dynamic Negative Guidance [6] address the separate problem of how much to extrapolate. None of these works provide quantitative evidence of disentanglement in structured attribute settings.

I wanted to ask: what if the baseline encoded everything except the target attribute?

Key Takeaway
- The baseline in CFG is a key design choice, rather than something trivial
- Replacing null with a targeted anchor achieved:
  - 45% lower KID at w=7 (>4σ)
  - 7.7% higher diversity at w=2
  - Eliminates mode collapse
  - Zero cost

Ref:
[1] Ho & Salimans (2022), Classifier-Free Diffusion Guidance, NeurIPS Workshop.
[2] Wu & De la Torre (2024), Contrastive Prompts Improve Disentanglement in Text-to-Image Diffusion Models, arXiv:2402.13490.
[3] Liu et al. (2022), Compositional Visual Generation with Composable Diffusion Models, ECCV.
[4] Sadat et al. (2025), Eliminating Oversaturation and Artifacts of High Guidance Scales in Diffusion Models, ICLR.
[5] Chung et al. (2025), CFG++: Manifold-constrained Classifier Free Guidance, ICLR.
[6] Koulischer et al. (2025), Dynamic Negative Guidance of Diffusion Models, ICLR.